Fire.Ai Project A - COO Memo

Recommendation
Select the base-stock policy for the current rollout. The forecast-driven candidate needs more tuning because it did not beat the baseline.
Final observed-sales proxy: proposed fill rate 92.2%; base-stock 94.1%.
Final simulated cost: proposed $103,538; base-stock $88,516.
95% target achieved? No - disclose shortfall and refine before rollout.

How the policy works
It uses empirical weekday forecast quantiles, lead-time protection, daily store transfer requests, Monday-only external POs,
and hard DC/store capacity limits. DC quantiles are computed from coherent demand scenarios, not summed marginal quantiles.
Observed sales are a proxy for demand: the source has no inventory or verified stockout labels. Recovery cases are sensitivities only.

Stress test
FOODS_3 doubled-lead-time disruption: fill rate 92.2%; warning mitigation 92.2%.
Mitigation pre-positions only during the permitted two-week warning and respects the same capacities.

Three risks and data needed
1. No inventory/stockout observations means censored demand is not identifiable. Collect on-hand,

availability and lost-sales signals.
2. Missing historical prices constrain cost accuracy. Collect listing-effective dates and complete price history.
3. Store demand dependence may differ from independent scenario assumptions. Add promotion, substitution and DC order data.
All figures are simulated policy outcomes, not observed business savings.